

Álvaro Ras Carmona, PhD

AI/ML scientist specialized in biology/life science

AI/ML scientist with 8 years of experience and a strong background in **artificial intelligence**, **data analysis**, and **biostatistics**. I have advanced programming skills and extensive experience working with large-scale biological datasets. My career has focused on the development and application of **machine learning models**, ranging from classical algorithms to advanced **deep learning** and **transformer architectures**, particularly encoder-only models, applied to biological problems. I have experience across the full model development lifecycle, including **data preparation**, **model training**, and **evaluation**. I am currently involved in a drug repurposing project leveraging artificial intelligence. Author of **16 scientific publications and book chapters**, including 9 as first or co-first author, as well as **14 contributions to national and international conferences**. I aim to leverage my technical and analytical expertise to contribute to the development of innovative solutions in the biotechnology and healthcare sectors.

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SKILLS

- **Machine Learning & Deep Learning:** Python · R · PyTorch · TensorFlow · Scikit-learn · Deep Learning · Transformers · Mixture of Experts (MoE) · MLPs · Large Language Models (LLM) · RAG · NLP · Protein Language Models (ESM-2, ProtBERT, ProtT5 ...) · Genetic Algorithms · Binary Classification · Cross-Validation y Model Evaluation · Hyperparameter Optimization with Optuna
- **Biología Computacional:** BioPython · Biostatistics · Protein Structure Prediction · AlphaFold3 · In Silico Peptide Design · BindCraft · BinderFlow · Protein-Protein Interaction (PPI) Prediction · AlphaPulldown · Computational chemistry · Scipion-Chem · Immunoinformatics
- **Análisis de Datos & Visualización:** NumPy · Pandas · SQL · UMAP · t-SNE
- **Computación Científica & Infraestructura:** HPC · GPU Computing · Slurm · Docker · Ollama · Linux · Bash · Git/GitHub

PROFESSIONAL EXPERIENCE

Computational Biology & AI Research Scientist

Jun 2025 - Present

Margarita Salas Center for Biological Research, Spanish National Research Council (CSIC), Madrid, Spain

- Contributed to the development of **artificial intelligence-based solutions** within a **project funded by Google** through the Silicon Valley Community Foundation.
- Designed and developed a Machine Learning model based on a **Mixture of Experts (MoE)** architecture comprising **multiple multilayer perceptrons (MLPs)** for **binary classification** tasks.
- Integrated **Transformer-based Protein Language Models** (ESM-2, ProtBERT, and ProtT5, among others) into Machine Learning pipelines to convert protein sequences into fixed-length vector representations (embeddings).
- Designed and implemented Python pipelines integrating **Retrieval-Augmented Generation (RAG)** techniques and **Large Language Models (LLMs)** via **Ollama** to predict the clinical success of therapeutic candidates in a drug repurposing project.
- Designed and generated in silico peptide candidates using computational protein design tools such as **BindCraft** and **BinderFlow**.
- Supervised and mentored a Master's student throughout the development of their research project during the 2025–2026 academic year.

International Experience: Predoctoral Research Stay

Sep 2022 – Dic 2022

Dipartimento di Scienze del Farmaco e della Salute, Università di Catania (Catania, Italia)

- Relocated to the University of Catania to collaborate on a project focused on predicting the next SARS-CoV-2 variant using **genetic algorithms**.

Computational Immunology Researcher

- Postdoctoral Researcher

Jul 2024 – Jun 2025

- Predoctoral Researcher

Oct 2019 – Jul 2024

Department of Immunology, Faculty of Medicine, Complutense University of Madrid, Spain

- Developed **machine learning models** for the classification of peptides as B-cell epitopes.
- Applied **dimensionality reduction techniques** (t-SNE and UMAP) for data analysis and visualization.
- Implemented and deployed predictive models on web servers to facilitate their use by the scientific community.
- Delivered 51 hours of practical immunology sessions to first-year medical students during the 2020–2021, 2021–2022, and 2022–2023 academic years, receiving highly positive feedback from students.

EDUCATION & TRAINING

Attendance at the AIHUB Summer School for Early-Career Researchers in AI Applied to Science

Sep 2026 (1 week)

AIHUB-CSIC

GPU-Based Parallel Supercomputing Workshop at HCP DRAGO

May 2026 (20 h)

Course organized by CSIC

PhD in Biomedical Research (Specialization in Bioinformatics) - *Cum Laude*

Oct 2019 – Jul 2024

Department of Immunology, Faculty of Medicine, Complutense University of Madrid, Spain

MSc in Omics Data Analysis (Bioinformatics)

Oct 2018 – Sep 2019

University of Vic – Central University of Catalonia, Barcelona, Spain

BSc in Biotechnology + University-Specific Qualification in Biotechnology Research Methodology

Sep 2014 – Jul 2018

Francisco de Vitoria University, Madrid, Spain